Joining, assembly & finishing

Higher · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe commercial assembly methods: adhesives, joints, knock-down fittings, fixings, snap and press fit, riveting, spot and arc welding, thermal bonding
- ☐ describe methods to identify assembly methods and the issues that influence them
- ☐ describe chrome plating, galvanising and plastic dip coating

I DO — Worked example

Read how a full-mark answer is built

1 Outline factors that influence the choice of assembly method for a product. (3 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Volume of production
- ✓ Materials being joined
- ✓ Whether the joint must be permanent or taken apart for maintenance

WE DO — Guided practice

Use the starters and word bank to help you

A flat-pack storage unit has a polypropylene drawer and MDF panels.

2 Describe how snap fits and knock-down fittings could be used in the assembly of the unit, giving an advantage of each. (4 marks)

Sentence starters

Snap fit: …
Snap fit advantage: …
Knock-down fitting: …
Knock-down advantage: …

Word bank

Snap fit · Knock-down fitting · consumer · MDF · cam lock · clicks · flexible · moulded

Tip: 4 marks = 4 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) What is the purpose of a pilot hole for a screw?
☐ To let the head sit flush ☐ To guide the screw and stop the wood splitting ☐ To make the screw removable ☐ To hold glue

b) A galvanised fence post can be identified by…
☐ A mirror finish ☐ A spangled, mottled surface ☐ A smooth coloured plastic coat ☐ Visible layer lines

c) Which joint is best for fixing shelves into the sides of a bookcase?
☐ Halving ☐ Housing ☐ Mortise and tenon ☐ Lap

d) Which method joins car body panels quickly on an automated line?
☐ Arc welding by hand ☐ Spot welding ☐ Dovetail joints ☐ Thermal bonding

4 Key terms: write the letter of the matching description next to each term.

1. **Mortise and tenon**	____	**A** Joins thin sheet from one side only.
2. **Housing joint**	____	**B** Hot metal dipped in plastic powder: protective, coloured, grippy finish.
3. **Knock-down fitting**	____	**C** Strong frame joint for tables, chairs and doors.
4. **Epoxy resin**	____	**D** Two-part adhesive that joins different materials.
5. **Pop rivet**	____	**E** Carcase joint for shelves in bookcases and cabinets.
6. **Dip coating**	____	**F** Non-permanent fitting for flat-pack furniture.

5 Explain what is meant by ‘Snap fit’. (2 marks)

6 Explain what is meant by ‘Spot welding’. (2 marks)

7 Explain what is meant by ‘Galvanising’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?